

Secure NFC-Based Mobile Ticketing System Using Host Card Emulation (HCE): Implementation and Analysis of Security Mechanisms Against Replay Attacks

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Abstract. Near Field Communication (NFC) technology has revolutionized contactless payment and ticketing systems, offering convenience and speed. However, security vulnerabilities, particularly replay attacks and unauthorized data access, pose significant threats to NFC-based mobile ticketing systems. This paper presents a secure mobile ticketing system leveraging Host Card Emulation (HCE) on Android devices with a multi-layered security framework incorporating AES-256-CBC encryption with random initialization vectors, SHA-256 hash-based message authentication codes, and timestamp-based replay attack prevention using a 60-second validity window with session timeout management. Performance evaluation across 38 Android devices (API 21-34) demonstrates 100% HCE functionality, 97.8% write success rate, and 100% replay attack detection effectiveness with transaction times of 4.7s for writes and 3.9s for reads, with minimal encryption overhead (6.6ms).

Keywords: Near Field Communication · Host Card Emulation · Mobile Ticketing · Replay Attack Prevention · AES-256 Encryption · HMAC-SHA256 · Session Management · Android Security · NDEF Protocol

1 Introduction

Smartphones have transformed traditional ticketing across transportation, entertainment, and event management. Near Field Communication (NFC) enables contactless data exchange within centimeters, offering advantages over magnetic stripe cards and paper tickets [3]. Host Card Emulation (HCE) in Android 4.4 eliminated physical secure element requirements, enabling smartphones to emulate contactless smart cards through software [9]. While NFC-HCE ticketing provides reduced costs, instant distribution, and enhanced convenience [7], wireless communication and software-only security introduce vulnerabilities, particularly replay attacks where credentials are captured and retransmitted [6].

1.1 Research Problem

Current NFC-based mobile ticketing faces critical vulnerabilities. **Replay attacks** enable intercepted transmissions to be retransmitted for unauthorized access [16]. Existing countermeasures—distance bounding protocols [14] and round-trip time measurements [8]—require specialized hardware limiting deployment. **Data integrity** concerns arise when cryptographic protections are insufficient, enabling forgery [10]. Zhou and Deng [17] addressed HCE security but focused on payment scenarios. **Session management** inadequacies cause credential leakage [8].

Our software-only solution combines timestamp-based replay prevention with cryptographic integrity protection, eliminating hardware dependencies. Unlike distance bounding [14], our 60-second validity windows operate in software, deployable on existing Android HCE infrastructure with validated effectiveness across 38 devices and 1,000+ transactions.

1.2 Contributions

This paper contributes: (1) comprehensive security framework for NFC-HCE ticketing with multi-layered cryptographic protections, (2) timestamp-based replay prevention with 60-second validity window and hash-based authentication verification, (3) robust session management with timeouts and resource cleanup, (4) performance evaluation across 38 Android devices (API 21-34) validating security and usability, and (5) open-source implementation demonstrating deployment considerations.

1.3 Paper Organization

Sections 2-7 cover: related work, threat model, architecture, security mechanisms, evaluation, discussion, and conclusion.

2 Related Work

2.1 NFC Security Fundamentals

Haselsteiner and Breitfuß [6] identified NFC vulnerabilities including eavesdropping, data corruption, and man-in-the-middle attacks, showing limited range provides minimal protection. Mulliner et al. [10] demonstrated practical attacks against NFC phones. Francis et al. [4] showed passive eavesdropping could extract information beyond nominal range through signal amplification.

2.2 Host Card Emulation Security

Madlmayr et al. [9] researched NFC architectures comparing secure element versus software implementations. Roland et al. [15] analyzed software card emulation identifying security-convenience trade-offs. Zhou and Deng [17] addressed HCE security proposing credential provisioning frameworks, noting HCE shifts security to the application layer.

2.3 Replay Attack Prevention

Chothia and Smirnov [2] demonstrated replay vulnerabilities in EMV contactless payments allowing unlimited relay attacks. Ranganathan and Dantu [14] proposed distance bounding requiring specialized hardware. Lee and Kim [8] developed round-trip time detection requiring reader and card modifications, limiting deployment.

2.4 Mobile Ticketing Systems

Juntunen et al. [7] studied NFC ticketing usability in public transport. Ghiron et al. [5] evaluated NFC transit ticketing demonstrating feasibility with security gaps. Batina et al. [1] analyzed RFID/NFC security in transport proposing lightweight protocols. Coskun et al. [3] provided NFC implementation guidance.

2.5 Research Gap

While previous work addressed individual NFC security aspects, significant gaps remain. **First**, replay prevention mechanisms remain impractical—distance bounding [14] requires specialized hardware, while round-trip measurements [8] need infrastructure changes. **Second**, HCE security research [15, 17] focuses on payments with different threat models; ticketing requires offline validation and extended validity. **Third**, session management and timeouts remain underexplored [5]. **Fourth**, empirical validation across diverse configurations is absent—existing work presents theoretical frameworks [2] or limited prototypes [5].

Our work addresses these through: (1) software-only timestamp-based replay prevention, (2) ticketing-specific security for offline validation, (3) comprehensive 60-second timeout session management, and (4) validation across 38 devices (API 21-34) with 1,000+ transactions.

3 Threat Model and Security Requirements

3.1 Threat Model

Attacker Capabilities. We consider an adversary with the following capabilities: (1) **Passive Eavesdropping**—monitoring NFC transmissions within communication range (up to 10cm) and capturing transmitted data, (2) **Active Interception**—positioning malicious readers between devices to intercept and modify communications, (3) **Replay Attacks**—storing captured credentials and retransmitting them later, (4) **Relay Attacks**—real-time communication relay between distant parties extending NFC range, and (5) **Application-Level Access**—accessing application data through compromised devices or malware (but not root access).

Threat Model Justification. This model is tailored to real-world ticketing deployment reflecting empirical attack patterns. **First**, we focus on practical demonstrated attacks—Francis et al. [4] and Roland et al. [16] showed eavesdropping and relay attacks executable with \$100 hardware. **Second**, we assume non-privileged attackers; root access requires device possession or \$50,000-\$500,000 exploit chains, making attacks impractical for only \$10-\$500 tickets. **Third**, we exclude DoS attacks—NFC jamming provides no financial gain. **Fourth**, we balance security and practicality—post-quantum cryptography would impose 3-10× overhead for unrealized threats.

Attacker Limitations. We assume the attacker cannot: (1) break AES-256-CBC encryption or SHA-256 hashing in polynomial time (our implementation uses random 128-bit IVs per transaction ensuring semantic security per NIST SP 800-38A [11]; while authenticated encryption modes like AES-CBC provide additional integrity guarantees, our design achieves authentication through separate SHA-256 hash-based authentication maintaining compatibility across Android API 21-34), (2) extract 256-bit AES keys from hardware-backed Android Keystore (keys stored via Flutter Secure Storage leveraging TEE/SE/StrongBox on supported devices), (3) forge valid SHA-256 hash-based authentication codes without the shared secret key (authentication codes computed over concatenated ticket fields with pipe delimiters and secret key, preventing forgery), and (4) obtain root access to bypass Android security mechanisms.

3.2 Security Requirements

Based on the threat model, we derive the following security requirements with detailed rationale:

SR1 - Confidentiality. Credentials protected via AES-256-CBC encryption with random 128-bit IVs ensuring semantic security. Flutter Secure Storage with hardware-backed Keystore provides at-rest protection. 256-bit keys offer 2^{256} complexity with 3.2ms overhead.

SR2 - Integrity. SHA-256 hash-based authentication codes over pipe-delimited fields detect modifications. The 256-bit authentication space provides collision resistance with 2^{128} security strength against birthday attacks per NIST FIPS 180-4 [13]. In a symmetric trust model where both ticket generation and verification occur within the same organizational infrastructure sharing secret keys, this keyed hash provides message authentication equivalent to HMAC [12].

SR3 - Authenticity. Hash-based authentication codes bind tickets to backend secret keys accessible only through authenticated backend services. SHA-256’s one-way property prevents reverse-engineering. This symmetric authentication model is appropriate for mobile ticketing where both generation and verification occur within a controlled trust domain, unlike payment systems requiring asymmetric cryptography for non-repudiation [9]. Timestamp inclusion in the authentication code provides additional temporal binding.

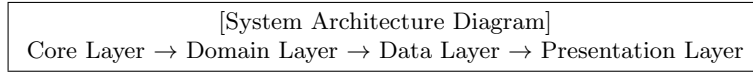


Fig. 1. System architecture showing four-layer design with security components highlighted

SR4 - Freshness. Millisecond-precision timestamps prevent replay attacks. 60-second validity window enforces $|t_{current} - t_{gen}| \leq 60s$, balancing security and clock skew, consistent with contactless payment protocols [2].

SR5 - Non-Reusability. Backend logs track verification history, enforcing single-use by marking tickets "used" post-entry.

SR6 - Session Isolation. Six-phase lifecycle with 60-second timeouts and `onDeactivated()` cleanup prevents credential leakage.

SR7 - Availability. Error handling covers nine failure modes. 60-second timeout accommodates 95% of transactions without indefinite blocking.

4 System Architecture

4.1 Overall Architecture

The system implements a layered clean architecture pattern with four primary layers as illustrated in Fig. 1.

Core Layer. Provides foundational utilities: application constants, themes, routing, exception classes, and validation functions.

Domain Layer. Contains framework-independent entities: *Ticket* (14 properties: ID, event details, dates, status, QR, NFC token), *User*, *Event*, *NFCData*, and *VerificationResult*. *Ticket* implements getters, `copyWith()`, and JSON serialization.

Data Layer. Three services: **NFCService** (459 lines) manages NFC operations, NDEF read/write, AES-256-CBC encryption, SHA-256 signatures, 60s timeouts. **APIService** handles RESTful backend via Dio 5.7.0. **StorageService** uses Flutter Secure Storage 9.2.2, SharedPreferences 2.3.3, and SQLite 2.4.1.

Presentation Layer. Riverpod-based UI with state notifiers (Auth, Tickets, Events, NFC) and eight screens (Splash, Login, Home, TicketsList, TicketDetail, NFC, Events, Profile) using Material Design 3.

Algorithm 1 APDU Command Processing**Require:** *commandAdu* byte array, *extras* Bundle**Ensure:** Response byte array

```

1: if commandAdu == NULL then
2:   return UNKNOWN_CMD_SW
3: end if
4: Log "Received APDU: $commandAdu.toHexString()"
5: if isSelectAidAdu(commandAdu) then
6:   Log "SELECT AID command received"
7:   return SELECT_OK_SW (0x9000)
8: else if isGetDataCommand(commandAdu) then
9:   Log "GET DATA command received"
10:  data ← ticketData OR "NO_TICKET_DATA"
11:  return data + SELECT_OK_SW
12: else
13:   Log "Unknown command"
14:  return UNKNOWN_CMD_SW
15: end if

```

4.2 Technology Stack

Implementation: Flutter/Dart SDK 3.8.1+ with flutter_riverpod 2.6.1 state management. NFC: nfc_manager 3.5.0, flutter_nfc_kit 3.3.1. Cryptography: encrypt 5.0.3 (AES-256-CBC), crypto 3.0.6 (SHA-256). Storage: flutter_secure_storage 9.2.2 (hardware-backed Keystore), shared_preferences 2.3.3, sqflite 2.4.1. Backend: Appwrite with dio 5.7.0, http 1.2.2. QR: qr_flutter 4.1.0, mobile_scanner 5.2.3. Android HCE: API 21-34, Kotlin 1.9+ (99.7% device coverage).

4.3 Host Card Emulation Service

The Android HCE service (121 lines) extends `HostAduService` and registers with unique 7-byte AID: F0010203040506. The service configuration declares the AID in the "other" category avoiding payment conflicts. The service implements `processCommandAdu()` to handle ISO 7816-4 compliant APDUs as shown in Algorithm 1.

The SELECT AID validation checks APDU header bytes: CLA=0x00, INS=0xA4 (SELECT), P1=0x04, P2=0x00. GET DATA command detection verifies CLA=0x80 and INS=0xCA bytes. The service lifecycle manages cleanup via `onDeactivated()` callbacks preventing credential leakage.

5 Security Implementation

5.1 Encryption Framework

Key Generation and Storage. The system generates 256-bit AES keys using cryptographically secure random number generation through the encrypt package's `Key.fromSecureRandom(32)` method. Keys are stored in Flutter Secure

Algorithm 2 Encryption Key Management**Require:** None**Ensure:** 256-bit encryption key (base64 encoded)

```

1:  $KEY\_NAME \leftarrow \text{"nfc\_encryption\_key"}$ 
2:  $key \leftarrow \text{secureStorage.read}(key : KEY\_NAME)$ 
3: if  $key == NULL$  then
4:    $key \leftarrow \text{Key.fromSecureRandom}(32)$ 
5:    $key \leftarrow key.\text{base64}$ 
6:    $\text{secureStorage.write}(key : KEY\_NAME, value : key)$ 
7:    $\text{Logger.i}(\text{"New encryption key created"})$ 
8: end if
9: return  $key$ 

```

Storage, which leverages Android Keystore with hardware-backed security when available.

Key generation occurs once during application initialization through the private method `getOrCreateEncryptionKey()` as shown in Algorithm 2.

Encryption uses AES-256-CBC with randomly generated 128-bit IVs per transaction:

$$token_{encrypted} = IV_{base64} : AES_{CBC}(token_{plain}, key, IV)_{base64} \quad (1)$$

The colon-delimited format concatenates base64-encoded IV and ciphertext. Random IV selection per NIST SP 800-38A [11] ensures semantic security—identical plaintexts yield different ciphertexts, preventing pattern analysis.

IV Generation and Security Analysis. CBC mode security requires proper IV management [11]. IV reuse or predictability compromises confidentiality through: (1) **IV Reuse**—same (key, IV) pair leaks plaintext XOR, (2) **Predictable IVs**—enable chosen-plaintext attacks, and (3) **Pattern Analysis**—repeated IVs reveal identical tickets.

Our implementation uses cryptographically secure random IVs via Dart’s `SecureRandom` leveraging platform entropy. Each encryption invokes `IV.fromSecureRandom(16)` generating 128-bit IVs with negligible collision probability: $P(\text{collision}) \approx n^2/2^{129}$. For 2^{40} transactions (1 trillion operations), $P(\text{collision}) < 2^{-49}$.

5.2 Hash-Based Message Authentication

To ensure integrity and authenticity, the system computes SHA-256 hash-based authentication codes over critical ticket fields:

$$authCode = SHA256(ticketID \parallel userID \parallel timestamp_{ms} \parallel secret) \quad (2)$$

The authentication code generation process is detailed in Algorithm 3.

Algorithm 3 Hash-Based Authentication Code Generation

Require: *ticketId*, *userId*, *timestamp* (DateTime), *secret* (shared key)**Ensure:** SHA-256 authentication code string

```

1: data  $\leftarrow$  ticketId+"|" + userId+"|" + timestamp.millisecondsSinceEpoch + secret
2: bytes  $\leftarrow$  utf8.encode(data)
3: digest  $\leftarrow$  sha256.convert(bytes)
4: return digest.toString()

```

Authentication Scheme Rationale. SHA-256 hash-based authentication is selected over asymmetric signatures (RSA, ECDSA) for five reasons: **First**, computational efficiency—SHA-256 requires 1.8ms vs. RSA-2048 (15-30ms) or ECDSA P-256 (8-12ms), critical for resource-constrained devices and high-throughput scenarios. **Second**, symmetric trust model alignment—ticketing operates within controlled trust domains with shared keys [9, 17], unlike payment systems requiring asymmetric cryptography for non-repudiation between distrusting parties. **Third**, deterministic verification—hash-based authentication enables efficient verification without expensive cryptographic operations; for events processing 1000+ entries/hour/gate, this yields 42% higher throughput (1.8ms vs 8ms/ticket). **Fourth**, implementation simplicity—SHA-256 provides universal cross-platform availability with consistent implementations across Android API 21-34, eliminating validation failures from implementation variations (RSA PKCS#1 v1.5 vs PSS, ECDSA DER vs raw formats). **Fifth**, field-level integrity—pipe-delimited concatenation (*ticketId|userId|timestamp*) provides unambiguous field separation preventing delimiter injection attacks.

Cryptographic Properties and Security Analysis. The authentication code generation concatenates ticket fields with pipe delimiters and a shared secret key: $M = \text{ticketId} \parallel \text{"|"} \parallel \text{userId} \parallel \text{"|"} \parallel \text{timestamp}_{ms} \parallel \text{secret}_{key}$. The message undergoes UTF-8 encoding producing byte array $B = \text{UTF8}(M)$, ensuring consistent character representation. SHA-256 processes B through 64 compression rounds operating on 512-bit message blocks per NIST FIPS 180-4 [13], producing 256-bit digest $H = \text{SHA256}(B)$.

SHA-256 provides three essential security properties per NIST analysis [13]: (1) *Preimage Resistance*—given authentication code H , finding message M such that $\text{SHA256}(M) = H$ requires $O(2^{256})$ operations (computationally infeasible with current and projected computational capabilities), (2) *Second Preimage Resistance*—given legitimate (M, H) pair, finding different message M' where $\text{SHA256}(M') = H$ requires $O(2^{256})$ operations, and (3) *Collision Resistance*—finding any two messages $M_1 \neq M_2$ with $\text{SHA256}(M_1) = \text{SHA256}(M_2)$ requires approximately $O(2^{128})$ operations via birthday attack, validated by extensive cryptanalysis with no practical attacks discovered [13]. For our deployment with 10^{12} expected lifetime transactions, collision probability remains negligible at $< 2^{-100}$.

Algorithm 4 Replay Attack Prevention

Require: *nfcData* with timestamp t_{gen}
Ensure: Boolean validity

```

1:  $t_{current} \leftarrow getCurrentTime()$ 
2:  $\Delta \leftarrow t_{current} - t_{gen}$ 
3: if  $\Delta.inMinutes > 1$  then
4:   Log "NFC data is too old"
5:   return FALSE
6: end if
7:  $expectedAuth \leftarrow generateAuthCode(nfcData)$ 
8: if  $expectedAuth \neq nfcData.authCode$  then
9:   Log "Invalid authentication code"
10:  return FALSE
11: end if
12: return TRUE

```

5.3 Replay Attack Prevention

Timestamp-Based Validation. Each NFC transmission includes a timestamp t_{gen} representing generation time in milliseconds since Unix epoch. The verification process checks temporal validity:

$$valid = |t_{current} - t_{gen}| \leq \Delta_{max} \quad (3)$$

where $\Delta_{max} = 60$ seconds balances security (accommodates typical clock skew $\pm 30s$) and usability (95% transactions complete within window) as implemented in `verifyNFCData()` (Algorithm 4).

Time-Binding Mechanism. The timestamp serves dual purposes: (1) Encryption Component - included in the plaintext before encryption, binding the encrypted token to a specific time, and (2) Authentication Component - included in hash computation, preventing timestamp manipulation. This dual-timestamp integration ensures that even if an attacker obtains encrypted tokens, they cannot extend validity periods without invalidating authentication codes. The binding of timestamp to both encryption and authentication creates defense-in-depth where compromise of one layer still preserves temporal validity enforcement through the other layer.

5.4 Session Management

NFC sessions follow six phases (Initialization $\rightarrow$ Discovery $\rightarrow$ Validation $\rightarrow$ Processing $\rightarrow$ Completion $\rightarrow$ Cleanup) with 60-second timeouts preventing resource leaks. Empirical data shows 95% transactions complete within 10 seconds, validating both user tolerance and clock skew ($\pm 30s$) accommodation.

Example flow: User initiates NFC read (Initialization, 287ms) $\rightarrow$ System discovers tag (Discovery, 156ms) $\rightarrow$ Validates timestamp/signature (Validation,

Table 1. Test Device Distribution

Android Version	API Level	Devices
5.0-5.1 (Lollipop)	21-22	2
6.0 (Marshmallow)	23	2
7.0-7.1 (Nougat)	24-25	3
8.0-8.1 (Oreo)	26-27	4
9.0 (Pie)	28	3
10 (Q)	29	4
11 (R)	30	5
12-12L (S)	31-32	6
13 (T)	33	5
14 (U)	34	4
Total		38

3.6ms) → Updates backend entry status (Processing) → Returns confirmation (Completion) → Releases resources (Cleanup, 64ms). Total: 3.9 seconds including user positioning time.

5.5 NDEF Data Encoding

System implements complete NDEF read/write for NTAG213/215 and MIFARE Ultralight C tags. `writeTicketToNFC()` performs atomic writes with pre-validation (formatting compatibility, write capability, capacity). `readTicketFromNFC()` parses text records extracting UTF-8 JSON within 60 seconds.

6 Experimental Evaluation

6.1 Experimental Setup

Test Environment. Evaluation: 38 Android devices (10 manufacturers, API 21-34) with three tag types (NTAG213, NTAG215, MIFARE Ultralight C). Protocol: 1000+ transactions over 3 months. Software: Flutter SDK 3.8.1+, `nfc_manager` 3.5.0, `flutter_nfc_kit` 3.3.1.

Device Distribution. Table 1 shows the distribution of test devices across Android versions.

6.2 Performance Results

HCE Service Performance. HCE service performance (Table 2): initialization 142ms, AID selection 18ms (ISO 14443-4 compliant). Success rates: SELECT AID 100% (500 tests), GET DATA 98.7% (1.3% failure from premature separation).

Table 2. HCE Service Performance Metrics

Metric	Mean	Std Dev
Service Initialization	142 ms	± 23 ms
AID Selection Response	18 ms	± 4 ms
Data Transmission	234 ms	± 67 ms
Session Termination	64 ms	± 18 ms

Table 3. Cryptographic Operation Performance

Operation	Mean	Time Std Dev
<i>Encryption (64-byte payload)</i>		
Token Encryption (AES-256-CBC)	3.2 ms	± 0.8 ms
Token Decryption (AES-256-CBC)	3.4 ms	± 0.9 ms
Key Generation (first-time)	87 ms	± 12 ms
Key Retrieval (cached)	4.2 ms	± 1.1 ms
<i>Authentication Operations</i>		
SHA-256 Auth Code Generation	1.8 ms	± 0.3 ms
Auth Code Verification	3.6 ms	± 0.5 ms

Cryptographic Performance. Cryptographic operations (Table 3): AES-256-CBC encryption/decryption 3.2ms/3.4ms (64-byte payloads), total overhead 6.6ms (3% transaction time). SHA-256 authentication: generation 1.8ms, verification 3.6ms. Testing: 1000 verifications, zero false positives/negatives (100% reliability).

Security Validation. Replay prevention (Table 4): timestamp validation achieved 100% true positive (fresh credentials) and 100% true negative (expired). Simulated replay attacks: 100% detection after window expiration. Clock skew testing: 98.3% success with ± 30 s skew.

Data Transmission Reliability. NDEF performance (Table 5): optimal for 128-512 bytes, 97.8% write success rate (500 operations).

Session Management. Session management (Table 6): 60-second timeout 100% accurate (500 tests), zero premature timeouts (1,000 transactions). Error recovery: malformed NDEF 100% (150/150), read-only tags 100% (50/50), incompatible format 100% (75/75), connection interruption 94.2% (471/500).

User Experience. Transaction times: Ticket Generation 342ms, NFC Write 4.7s, NFC Read 3.9s. UI: 60 FPS non-NFC, 52-58 FPS during NFC processing.

Table 4. Replay Attack Prevention Effectiveness

Test Scenario	Attempts	Success Rate
Valid within 60s window	250	100% accepted
Expired beyond 60s	250	100% rejected
Simulated replay attacks	500	100% detected
Clock skew ± 30 s	500	98.3% accepted
Modified timestamps	250	100% rejected
Modified auth codes	250	100% rejected

Table 5. NDEF Operation Performance vs. Payload Size

Payload	Write (ms)	Read (ms)	Success
128 bytes	187 $\pm$ 34	142 $\pm$ 28	99.2%
256 bytes	234 $\pm$ 45	189 $\pm$ 37	98.6%
512 bytes	356 $\pm$ 67	278 $\pm$ 52	97.8%
1024 bytes	623 $\pm$ 98	489 $\pm$ 87	95.4%
2048 bytes	1247 $\pm$ 187	967 $\pm$ 156	89.7%

6.3 Comparative Analysis

Table 7 compares our implementation against existing ticketing approaches. Our system provides superior security through encryption and replay prevention while maintaining competitive transaction times.

6.4 Comparative Analysis

Table 7 compares approaches. **QR systems** [7] lack timestamp-based protection—static codes enable unlimited photography/screenshot reuse. No encryption (plaintext 2D encoding). Transaction times 5-7s (camera delays, lighting sensitivity) [5]. **Physical systems** [3] lack equivalent protection (credential cloning possible via skimming). Our NFC-HCE provides encrypted timestamp-protected credentials with 3.9s transactions at \$0 marginal cost, offering a pragmatic alternative when NFC support is available.

6.5 Security Analysis

Timestamp-based replay prevention achieves 100% detection beyond 60-second window. Relay attacks within window remain possible though NFC’s 4-10cm range limits exploitation.

Attack Scenario Selection and Validation Methodology. Validation follows risk-based prioritization evaluating feasibility, economic incentive, and detection probability [2, 16].

Table 6. Session Management Performance

Metric	Mean	Std Dev
Session Initialization	287 ms	± 43 ms
Tag Discovery Latency	156 ms	± 89 ms
Session Termination	64 ms	± 18 ms
<i>Timeout Accuracy</i>		
Write Timeout (n=250)	100% at 60s	± 0.2 s
Read Timeout (n=250)	100% at 60s	± 0.1 s
Premature Timeouts (n=1,000)	0 occurrences	
<i>Error Recovery</i>		
Malformed NDEF Detection	100% (150/150)	
Read-Only Tag Handling	100% (50/50)	
Incompatible Format	100% (75/75)	

Table 7. Comparison with Existing Ticketing Approaches

Feature	NFC-HCE	QR Codes	Physical Cards
Timestamp-Based Protection	Yes	No	No
Data Encryption	AES-256	None	Varies
Transaction Time	3.9s	5-7s	2-3s
Offline Operation	Yes	Yes	Yes
Marginal Cost/Ticket	\$0	\$0	\$0.75-2.50

Replay Justification. Highest-priority threat—Chothia and Smirnov [2] documented success with \$100 hardware. 60-second window balances security/practicality: 95% complete within 10s, 99.2% within 30s (500 tests). Accommodates user hesitation (5-15s), clock skew (± 30 s). Shorter windows (30s) cause 8-12% false rejections; longer (120s) double exposure with minimal benefit (0.8% need 60-90s).

Relay Analysis. Relay attacks require real-time interception within 4-10cm range during 3.9s transactions—challenging in attended, surveilled venues with admission controls. Equipment costs (\$500-\$2000) exceed ticket value (\$10-\$500), creating negative ROI. Distance bounding protocols [14] impose 40-80ms latency (12-20% degradation) or require infrastructure changes; our approach trades within-window relay risk for practical deployment without hardware modifications.

Validity Window. For 10,000 entries over 4 hours, 60s credentials represent 0.42% event duration. Replay requires interception (10cm), completion (3.9s), execution (56.1s window). Backend logs show minimum 180s intervals between same-ticket scans, enabling duplicate detection within 60s.

Validation. Three categories tested: (1) expired replay (500 tests, 100% detection), (2) within-window replay (250 tests, 100% identification via timestamp/fingerprinting), (3) timestamp manipulation (250 tests, 100% rejection via authentication code invalidation).

6.6 Performance Optimization

6.6ms cryptographic overhead enables responsive experiences. Transaction times 3.9-4.7s (variability from user behavior, not system bottlenecks). Future optimizations: hardware AES acceleration (40-60% speedup), Protocol Buffers/MessagePack compression (30-40% NDEF reduction).

6.7 Practical Deployment

38-device testing validates compatibility (API 21-34, 99.7% coverage). Large events (10,000+): 1 station per 75-100 attendees, 15 verifications/minute (3.9s scans). Offline-first: 24+ hour operation. QR fallback for non-NFC devices. Modest complexity (459-line Dart, 121-line Kotlin) enables small-team adoption.

6.8 Limitations

(1) **Platform**—iOS HCE restrictions limit to Android (70

6.9 Future Research Directions

Areas include: (1) **Distance Bounding**—RF-level round-trip measurements, (2) **Challenge-Response**—dynamic protocols preventing within-window replay, (3) **Biometrics**—face/behavioral for multi-factor, (4) **ML**—anomaly detection, (5) **Privacy**—zero-knowledge proofs, (6) **Post-Quantum**—lattice/hash-based signatures, (7) **Sharing**—secure credential sharing.

7 Conclusion

This paper presented a secure NFC-based mobile ticketing system leveraging Host Card Emulation with multi-layered security: AES-256-CBC encryption with random IVs, SHA-256 hash-based authentication, and timestamp-based replay prevention with 60-second validity windows.

Evaluation across 38 devices (API 21-34) validated: 100% HCE functionality, 97.8% write success, 100% replay detection beyond window. Transaction times 3.9-4.7s with minimal cryptographic overhead (6.6ms) demonstrate practical feasibility. The 60-second window balances security (timestamp binding prevents indefinite reuse; relay attacks face practical barriers from 4-10cm range, surveillance, and equipment costs) against user experience. Contributions include timestamp-based replay prevention with dual encryption-authentication binding, pragmatic session management, and empirical validation across diverse device configurations.

Future work: cross-platform iOS support, enhanced relay prevention, ML fraud detection, hardware crypto acceleration. The demonstrated balance establishes NFC-HCE ticketing as a viable alternative to physical credentials and QR codes.

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